

4. (25 marks) Consider the following quadratic form

$$Q(\mathbf{x}) = abx_1^2 + bx_2^2 + ax_3^2 + 2bx_2x_3.$$

- (a) (4 marks) Identify the *symmetric matrix of coefficients* A associated with this quadratic form. That is

$$Q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}.$$

Find all the principal submatrices of A .

- (b) (6 marks) Define what it means for the quadratic form $Q(\mathbf{x})$ to be *negative definite*. How would you be able to assess, in terms of the properties of A , whether $Q(\mathbf{x})$ is negative definite?
- (c) (15 marks) Find the values of the parameters a and b such that the quadratic form $Q(\mathbf{x})$ is respectively *indefinite*, *positive definite* and *negative definite*. Then, plot these regions in $\mathbb{R}^2$.

5. (25 marks) Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a *differentiable convex function*.

- (a) (6 marks) State the definitions of a *critical point* and of a *global minimizer* for the function f .
- (b) (10 marks) State the definition of *convex function* for the function f . Then, construct an equivalent definition of *convexity for a function f that is differentiable, but not twice differentiable*.
- (c) (9 marks) By exploiting the previous definition, show that any *critical point* of f is a *global minimizer* of f .

6. (50 marks) For $a > 0$, consider the problem:

$$\begin{aligned} \max_{x,y} \quad & ax + \log y \\ \text{s.t.} \quad & 2x + 2y \leq 7 \\ & y \geq a, x \geq 0. \end{aligned}$$

- (a) (8 marks) Establish whether the objective function and the constraints are respectively concave, convex, or neither. Can you solve this optimization problem using the Kuhn-Tucker method? That is, for what values of $a > 0$ is the Slater condition satisfied?
- (b) (8 marks) Suppose the Slater condition is satisfied, write down the Kuhn-Tucker necessary first order conditions that must be satisfied by the solution of the constrained optimization problem. Are solutions to these conditions also maximizers of the Lagrangean?
- (c) (20 marks) Suppose the Slater condition is satisfied, solve the constrained optimization problem in terms of a .

- (d) (10 marks) State the Envelope Theorem and use it to find the slope of value function with respect to the parameter a for $a \in [0, 3]$.
- (e) (4 marks) For what value of $a \in [0, 3]$ is the value function maximized?